Facebook Marketplace

Live Sellers — Data Analysis

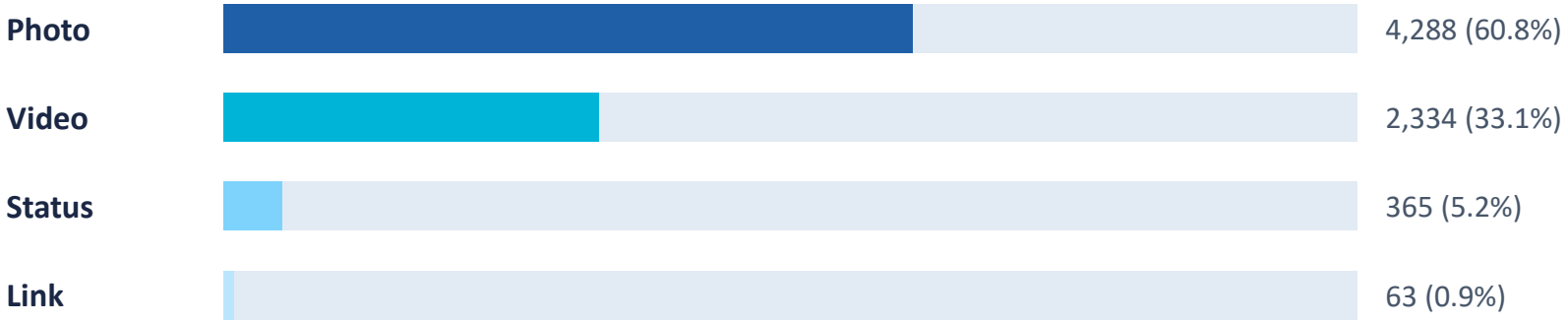
Exploring engagement patterns, correlations, and clustering insights

Dataset: 7,050 posts | Period: 2012–2018 | Source: UCI Machine Learning Repository

Dataset Overview

7,050	16	2012–2018	4 Types
Total Posts	Columns	Date Range	Post Types

Post Type Distribution



Q1: How Does Upload Time Affect Reactions?

Hourly Analysis

Peak engagement occurs between 18:00–20:00 (6–8 PM), with average reactions exceeding 430. Early morning hours (2–4 AM) show the lowest engagement (~160 reactions).

Daily Analysis

Thursday drives the highest reactions (~262 avg), followed by Wednesday and Sunday. Monday and Tuesday see noticeably lower engagement, dipping below 210.

Yearly Analysis

Reactions peaked dramatically in 2015 (~1,550 avg), then declined sharply. 2016–2018 show a stabilized but much lower baseline (~100–200), reflecting platform shifts.

📌 **Key Takeaway: Post in the evening on Thursdays for maximum reach — timing makes a measurable difference in engagement.**

Q2: Correlation Between Reactions, Comments & Shares

Method: Pearson Correlation Coefficient — measures linear relationship strength and direction between engagement metrics.

Reactions vs. Comments

0.15

Weak • Positive

Posts that receive more reactions only slightly tend to attract more comments. The relationship is present but not reliable.

Reactions vs. Shares

0.25

Weak • Positive

A slightly stronger link exists between reactions and shares. Popular posts are somewhat more likely to be shared, but the effect is modest.

Comments vs. Shares

0.64

Strong • Positive

The strongest finding: posts that generate discussion (comments) are significantly more likely to be shared. This suggests conversation drives virality.

📌 **Key Takeaway: Drive comments to amplify shares. The comments–shares link (r=0.64) is the most actionable insight for boosting content virality.**

Q3: K-Means Clustering (K = 3)

01

Feature Selection

Selected 10 features: status_type, num_reactions, num_comments, num_shares, num_likes, num_loves, num_wows, num_hahas, num_sads, num_angrys.

02

OneHot Encoding

status_type is categorical (photo, video, link, status). Applied OneHotEncoder via ColumnTransformer, with numerical columns passed through unchanged.

03

Feature Scaling

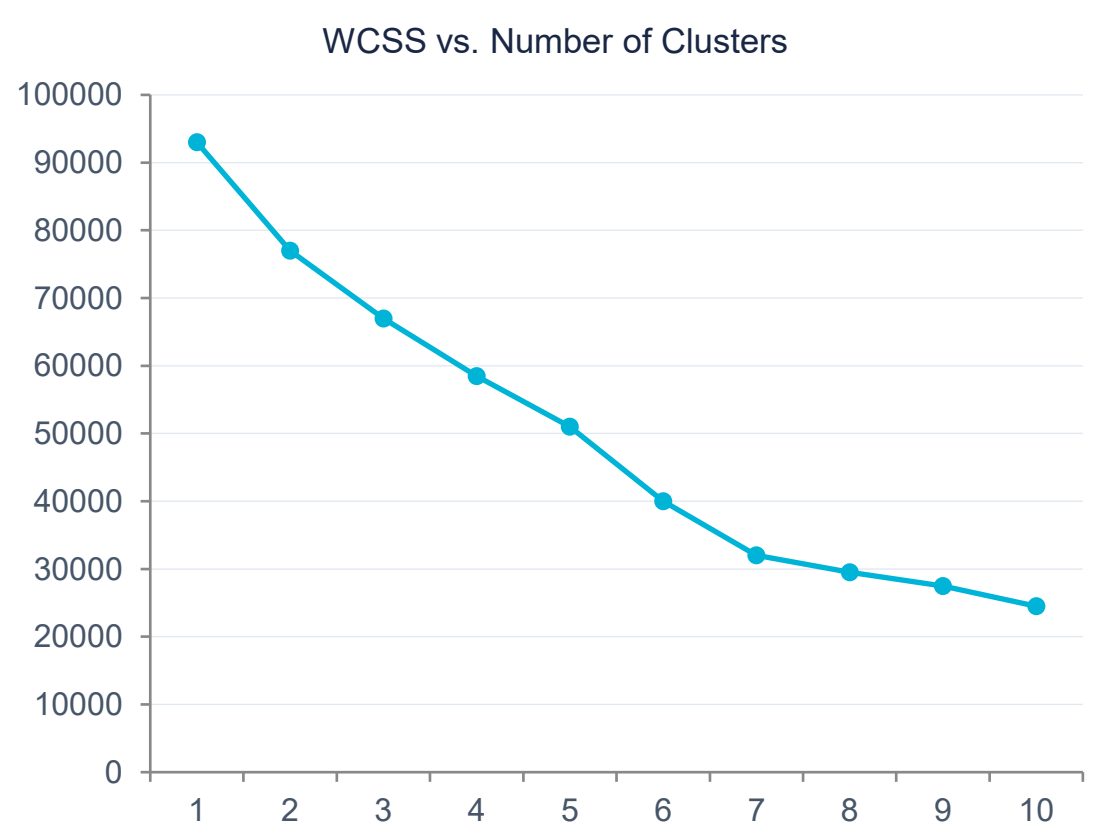
Standardized all features using StandardScaler to ensure no single metric dominates due to scale differences.

04

Model Training

Trained KMeans with K=3 (n_clusters=3, random_state=42). Cluster labels were appended to the original dataframe for analysis.

Q4: Optimal Clusters via the Elbow Method



What is the Elbow Method?

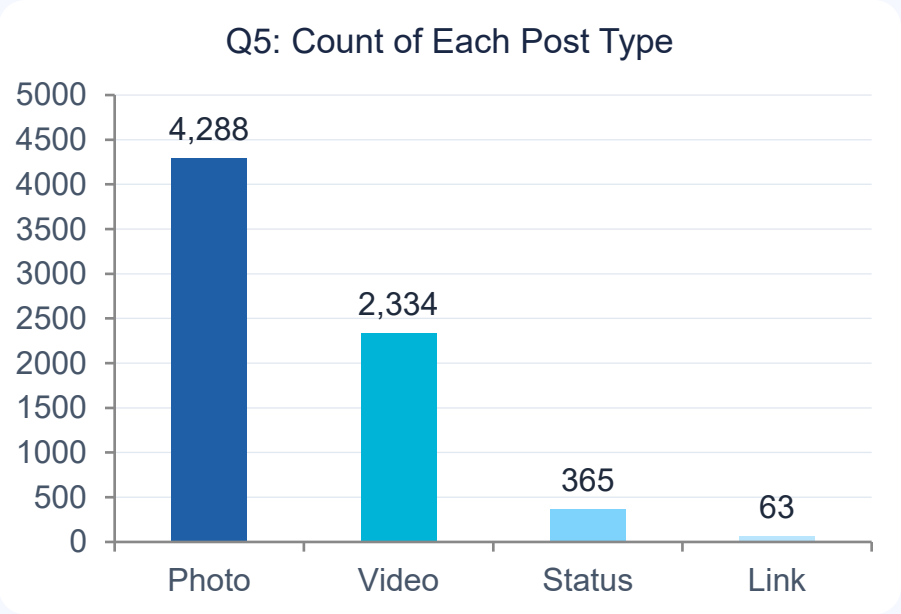
The Elbow Method helps find the optimal K by plotting the Within-Cluster Sum of Squares (WCSS) for each value of K.

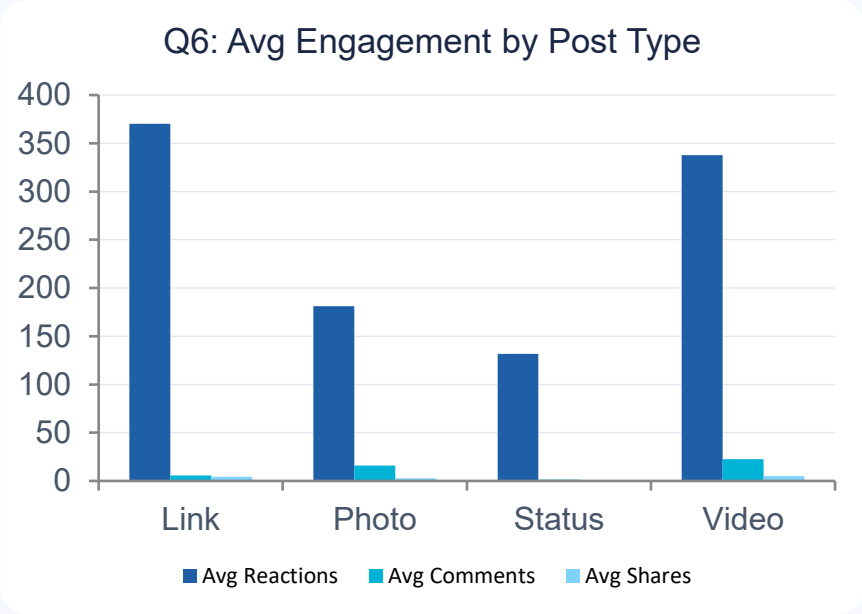
As K increases, WCSS decreases — but the rate of improvement slows. The 'elbow' point marks where adding more clusters yields diminishing returns.

In this analysis, the curve bends notably around K = 6, making it the optimal number of clusters for this dataset.

📌 **Optimal K = 6 identified from the elbow — used to retrain the final KMeans model with k-means++ initialization.**

Q5 & Q6: Post Type Counts and Average Engagement





Photo

Most frequent (60.8%), but mid-range reactions. Volume is its strength.

Video

Strong across all metrics — highest average comments & shares among common types.

Link

Fewest posts, but highest average reactions. Rare but punchy.

Status

Text-only posts show very low engagement across all metrics.

Key Takeaways

⏰ Timing Matters

Evening posts (6–8 PM) on Thursdays consistently outperform other slots. Year-on-year patterns also show engagement peaks in 2015.

🔗 Comments Drive Shares

The strongest correlation (r=0.64) is between comments and shares — not reactions. Sparking conversation is the real key to virality.

🎯 6 Clusters Emerge

The elbow method identifies K=6 as optimal, revealing meaningful audience segments beyond the obvious photo/video split.

📸 Content Type Counts

Photos dominate in volume, but link posts generate the highest per-post reactions. Video excels at driving comments and shares.

Thank You!

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Questions? Feedback? Let's discuss!